Model Performance Across Species

Four models, three evaluation regimes, from in-distribution discrimination to genome-scale deployment

August 2026 (0.6-clustering retrain)

1 Overview

Historical August 2026 comparison of ProkBERT, NT-LoRA, CNN-v1 and CNN-v2 after 0.6-identity clustering. These are earlier checkpoints, not the later deployed Evo2 trio. Tables preserve reported measurements; original metric/split/track inputs are not committed with this report.

2 Regime A: species discrimination

Species-labelled positives are compared with the same heterogeneous negative pool. The plain-0.6 test has 967/3,318 positives, not the deployed 547/3,053. AP baselines are about 0.29 and 0.18. Source labels are not uniformly assay-confirmed.

Table 1: AUROC on the T15 test set, by species, against the shared heterogeneous negative pool. (n = source-labelled positives in the test split, not uniformly assay-confirmed; **bold** = highest point estimate per species.)

Species (n_{pos})	ProkBERT	NT-LoRA	CNN	CNN-v2
E. coli (404)	**0.933**	0.929	0.878	0.920
C. beijerinckii (360)	**0.947**	0.924	0.914	0.936
C. acetobutylicum (166)	**0.946**	0.921	0.908	0.912
B. subtilis (37)	0.897	**0.936**	0.885	0.884

ProkBERT leads point estimates on E. coli and both Clostridium sources; NT leads B. subtilis. CNN-v2 exceeds CNN-v1 on three species. Fixed-0.5 recall differs from discrimination because raw score scales are uncalibrated; no architecture effect or significant source difference is isolated.

3 Regime B: composition controls

The 967 positives and 611 SRS windows test sequence structure beyond the supplied composition control, not a unique causal motif mechanism. Custom shuffles need per-sequence QC for exact preservation.

Table 2: Test positives (967) against SRS dinucleotide-shuffle negatives (611 windows). Composition is controlled within the shuffled source sequences, not guaranteed exactly matched across the filtered pools. AUROC with 90% bootstrap CI; chance = 0.5. **Bold** = best.

Model	AUROC (positives vs SRS shuffle) [90% CI]
ProkBERT	0.906 [0.90–0.91]
NT-LoRA	0.896 [0.89–0.91]
CNN	0.896 [0.89–0.91]
CNN-v2	**0.924 [0.92–0.93]**

CNN-v2 leads at 0.924. Other negative classes are different contrasts; CDS AUROC is 0.925–0.980. T9 uses 68 in-genus and 25 out-of-genus positives against shuffles. In-genus AUROC is ProkBERT 0.736

[0.65–0.80], CNN-v2 0.659 [0.57–0.74], CNN-v1 0.629 [0.54–0.71] and NT 0.440 [0.36–0.53], versus earlier NT 0.71. Out-of-genus none clears chance and ProkBERT records 0.17. These small library-specific samples do not establish a general genus boundary.

4 Regime C: published-TSS ranking

C. ljungdahlii supplies 1,245 primary TSS. Historical full/restricted tracks contain 9.26M/1.77M windows; restriction covers 19.2% of windows and excludes 11 reachable starts, limiting recall to 99.1%. TSS/window density is not promoter prevalence.

Coverage uses same-strand 81-bp windows expanded by 40 bp. Unmatched starts merge at gaps at most 81 bp; the historical mask includes exact endpoint touches. Precision proxy is covered TSS/(covered TSS+unmatched peaks). This is not biological purity, and the strain supplied silver/negative training windows.

Table 3: Historical *C. ljungdahlii* spatial benchmark (both strands; known-TSS/window density 0.013% full, 0.070% restricted). Precision proxy at four published-TSS recall targets and benchmark-unmatched peaks (FP) at 50% recall. **Bold** = best within each scan. Neither FP nor the precision proxy establishes biological error or purity.

		Precision @ recall target				FP peaks
Model	Scan	0.1	0.5	0.7	0.9	@0.5
ProkBERT	full	23.6%	18.5%	15.8%	11.5%	2,753
NT-LoRA	full	18.4%	13.1%	10.8%	7.9%	4,139
CNN	full	**58.3%**	**32.7%**	**23.9%**	**12.7%**	**1,284**
CNN-v2	full	50.8%	31.9%	22.6%	11.5%	1,332
ProkBERT	restrict	38.6%	33.1%	29.6%	24.3%	1,257
NT-LoRA	restrict	51.7%	32.5%	27.9%	22.5%	1,294
CNN	restrict	**63.0%**	39.2%	33.7%	24.9%	967
CNN-v2	**restrict**	59.8%	**44.3%**	**36.8%**	**26.5%**	**782**

CNN-v2 leads restricted recall 0.5–0.9; CNN-v1 leads at 0.1. Restriction changes unmatched-peak counts, not verified coding-sequence false positives. Earlier ProkBERT re-scan records about 45%/760 unmatched peaks at recall 0.5, versus 33.1%/1,257 after retraining, while validation AP changes 0.876 to 0.878. Clustering and hyperparameters both changed, so neither cause is isolated. Historical CNN-v2 0.9/0.6 comparisons (40%/44%) likewise do not prove a motif or leakage mechanism.

5 What we learned

Curated AP does not determine spatial ranking. The proposed four-model ensemble was not measured in this report; the later trio is separate. CNN-v2's raw negative rejection 0.766 at threshold 0.5 illustrates why reliability and ranking need separate evaluation. Only strictly increasing score-only transforms preserve ranks/ties; AT maps can reorder them.

Cluster grouping, SRS controls and incomplete TSS truth retain their limits. Deployed training still contains the wrong-assembly B. licheniformis positives. No target activity, sigma use, constitutive expression or cotranscription is established. Current interpretation is in docs/RESULTS.md.